JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 95, NO. B4, PAGES 4579–4585, APRIL 10, 1990

Bathozonal Tilt Corrections to Paleomagnetic Data From Mid-Cretaceous Plutonic Rocks: Examples From the Omineca Belt, British Columbia

E. IRVING

Pacific Geoscience Centre, Geological Survey of Canada, Sidney, British Columbia

D.A. ARCHIBALD

Geological Sciences, Queens University, Kingston, Ontario, Canada

The Skelly Creek batholith (49.4°N,116.6°W, 94-99 Ma) and Summit stock (49.1°N,117.0°W, 102 Ma) are intruded into the southern Omineca Belt of southeastern British Columbia. Bathozonal mineral assemblages from contact aureoles of these plutons and others of the same age indicate that post emplacement tilting of the Purcell anticlinorium has probably not exceeded about 5°, whereas the Summit Stock has been tilted 24° down to the west. The Skelly Creek batholith has a mean direction of magnetization (*D,I*) with respect to present horizontal 349°,74° (13 sites, k = 53, α_{95} = 6°). The mean direction of magnetization of the Summit stock is 48°,72° (nine sites, k = 290, α_{95} = 3°), and after correction for a westerly dip of 24°, the mean direction becomes 318°,74°. Directions of the Skelly Creek batholith and tilt-corrected direction from the Summit stock are in agreement with the reference field for cratonic North America. Evidently, this part of the eastern Omineca Belt was close to the margin of ancestral North America in mid-Cretaceous time, essentially where it is today.

INTRODUCTION

Paleomagnetic directions (paleodirections) from mid-Cretaceous (about 100 Ma) rocks of the Western Cordillera in British Columbia and northern Washington are inconsistent with data from Cretaceous rocks of the North American craton. These aberrant data are mainly from large plutons which contain no internal evidence of the attitude of paleohorizontal at the time magnetization was acquired. Hence the discordances have been interpreted as indicative either of previously unrecognized systematic geological tilts of about 30° to the southwest over a large region or of displacements of over 2000 km from the south and clockwise rotations of about 60° [*Beck and Noson*, 1972; *Beck et al.*, 1981; *Irving et al.*, 1985]. *Gabrielse* [1985] has recognized structures in north central British Columbia along which as much as 900 km of right-lateral displacement could have occurred, and *Marquis and Globerman* [1988], have observed paleomagnetically 1500 ± 950 km of northward movement from bedded volcanics of the Carmacks Group (70 Ma) of the Whitehorse Trough, so that although there are some supporting reasons for supposing that displacements of up to 1000 km might have occurred, there are apparently none for displacements in excess of 2000 km, as indicated by mid-Cretaceous data from intrusions. Indeed, *Price and Carmichael* [1986] have argued strongly that such large displacements did not occur. The dilemma posed by the mid-Cretaceous paleomagnetic data (tilt or translation?) is a major unresolved problem in Cordilleran tectonics.

One question of key importance in this discussion is, how much nearer to the craton do these enigmatic aberrancies extend? The rock unit closest to the craton that is known to have aberrant magnetizations is the Axelgold intrusion (AX of Fig. 1), a body that is intruded into Cache Creek Terrane, the most easterly element of the Intermontane Belt [*Monger and Irving*, 1980; *Armstrong et al.*, 1985]. The purpose of this paper is to report data from the Omineca Belt, the belt immediately to the east of the Intermontane Belt (Figure 1).

The southeastern Omineca belt, which includes the Kootenay Arc and the Purcell Anticlinorium, is composed mainly of late Precambrian and lower Paleozoic sedimentary rocks, which are interpreted to have been part of the miogeoclinal margin of ancestral North America [*Price*, 1981]. Therefore, to observe concordance between paleomagnetic data from Cretaceous rocks of this region and the craton would be a reasonable expectation and would provide evidence in support of the geological interpretation of proximity.

Ideally, in such work, well-dated Cretaceous bedded sequences should be studied, so that directions of magnetizations can be obtained relative to the original paleohorizontal, from which the paleopole and displacements can be calculated. In the southeastern Omineca Belt such young bedded sequences do not exist, and in order to make tests, one is obliged to sample intrusive rocks, which, as already noted, do not themselves contain evidence of the attitude of the paleohorizontal at the time magnetization was acquired. However, the problem

Copyright 1990 by the American Geophysical Union

Paper number 89JB02986.
0148-0227/90/89JB-02896$05.00

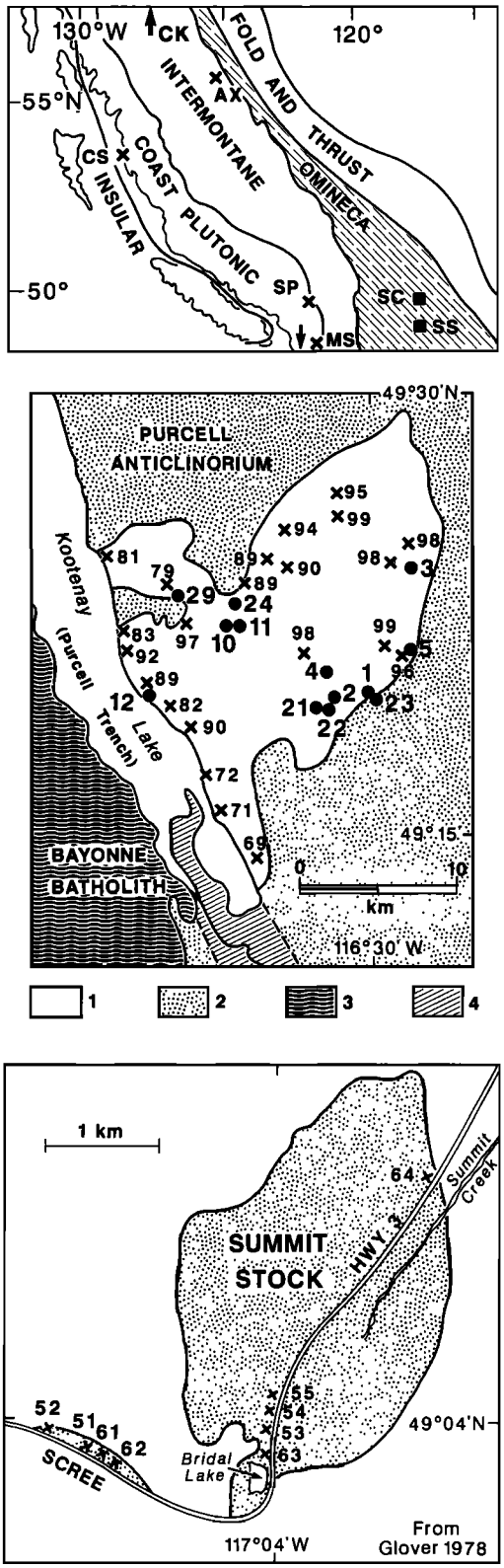


Fig. 1. Tectonic belts of the Canadian Cordillera above: SC and SS are the Skelly Creek batholith and Summit stock; AX, Axelgold intrusion 125-115 Ma [*Monger and Irving*, 1980; *Armstrong et al.*, 1985]; CK, Carmacks Group of the Yukon and northern British Columbia; CS, Captains Cove, Stephen, and Gil Island plutons, 109-102 Ma [*Symons*, 1977]; MS, Mount Stewart batholith, 100 Ma [*Beck et al.*, 1981]; SP, Spuzzum pluton, 100 Ma [*Irving et al.*, 1985]. Skelly Creek batholith middle: crosses are K/Ar biotite ages in Ma from *Archibald et al.* [1984], bold dots are paleomagnetic sampling sites; 1, the batholith, 2, Purcell Anticlorium, 3, Bayonne batholith, 4, Quaternary alluvium. Summit Stock below; crosses are sampling sites.

is mitigated by the possibility, exploited in this paper, of estimating regional post emplacement tilt using metamorphic mineral assemblages occurring in contact aureoles of plutons of similar age. The mineral assemblages we have compiled are those listed by *Carmichael* [1978]. Locations of most of the assemblages were presented by *Archibald et al.* [1984]. Additional mineral assemblage data for the Angus Creek stock, Sawyer Creek stock, and the eastern lobe of the Fry Creek batholith were collected during the course of the present work. These new data support the original conclusion of *Archibald et al.* [1984] concerning the depth of emplacement of the mid-Cretaceous plutons. If the tilt corrections estimated by these procedures and the foregoing geological interpretation are correct, then the paleodirections should be concordant with those observed in coeval rocks from the North American craton.

Paleomagnetic measurements were made on Schonstedt spinner magnetometers. Stepwise thermal and alternating field (AF) demagnetization was carried out using a Schonstedt TSD-1 furnace and a GSD-5 tumbling demagnetizer. All cores were oriented by sun and magnetic compasses.

TILT OF SKELLY CREEK BATHOLITH

The Skelly Creek batholith (Figure 1) is one of a suite of granitic stocks and batholiths which were emplaced into the southern Omineca Belt. Like the others, it was intruded after regional penetrative deformation was complete and has a narrow, contact aureole. It is intruded into the Purcell Anticlinorium. The most common rock type is a coarse-grained, K-feldspar-megacrystic, hornblende-biotite granodiorite. Mafic and metasedimentary xenoliths sometimes occur.

Considering the suite of plutons as a whole, the oldest K/Ar biotite dates range between 95 and 100 Ma [*Archibald et al.*, 1984]. *Hoy and Van der Heyden* [1988] have reported a U-Pb zircon date of 94 Ma for a small pluton to the east of the study area. Evidently, the suite of plutons was emplaced over a short period in mid-Cretaceous time. Numerous K/Ar ages are available from the Skelly Creek batholith itself, with maximum values between 94 and 99 Ma and younger ages ranging down to 69 Ma along the Purcell Trench (Figure 1). The latter were interpreted by *Archibald et al.* [1984] to be the result of slow cooling followed by rapid uplift. However, it is also possible that they reflect a thermal pulse in Eocene time, because the batholith is near the eastern margin of the zone of mid-Eocene heating, uplift and extension in the southern Omineca Belt [*Parrish et al.*, 1988]. In order to minimize the magnetic effect of this later thermal event, samples were obtained at or near those places from which the oldest K/Ar biotite ages were obtained.

Although not common, mineral assemblages useful for estimating the depth of emplacement occur in the contact aureoles of many of this suite of Cretaceous plutons [*Archibald et al.*, 1984]. Inner

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

parts of aureoles are characterized by andalusite, sillimanite, garnet, and, locally, staurolite with coexisting quartz, muscovite, and biotite, for example, in the White Creek batholith [*Reesor*, 1958]. Kyanite is present only in the aureole of the Fry Creek batholith, where it is synkinematic and related to the much older mid-Jurassic regional metamorphism of the host rocks. In the absence of sillimanite, andalusite does not occur with K-feldspar. These data indicate that the plutons of the Purcell Anticlinorium were emplaced in bathozones 2 to 3 of *Carmichael* [1978] and have been uplifted subsequently by about 15 km. The plutons are postkinematic, and many were emplaced across major reverse faults that contain no evidence of post-Cretaceous movement. Therefore, for the following calculations we assume that subsequent to mid-Cretaceous time, the Purcell Anticlinorium behaved as a quasi-rigid block or sheet. Bathozonal data were collected over an east-west distance of about 60 km and a north-south distance of about 120 km from a region centred on the Skelly Creek batholith. Using a conversion factor of 3.7 km/kbar and assigning an error of 0.5 kbar to the pressure of the invariant points used to define the transitions from bathozone 1 to 2 and bathozone 3 to 4, a rough estimate of the tilt can be made. On this basis, the tilt in an east-west sense (about a N-S axis), of the mid-Cretaceous paleohorizontal is estimated to be 5 ± 3° either side of present horizontal. The tilt either side of present horizontal in a north-south sense is 2 ± 1°. These estimates are rough, but they indicate that tilting is unlikely to have exceeded a few degrees.

PALEOMAGNETISM OF SKELLY CREEK BATHOLITH

Although massive in outcrop, the body contains a network of fine fractures, making sampling tedious and difficult. Fractures are often reddened. Between three and nine drill cores were taken at each site and were oriented by magnetic and sun compasses.

The remanent magnetization is of variable stability, and the granodiorite itself is not generally a good recorder of the paleofield. However, preliminary sampling showed that mafic (diorite) xenoliths, some metasedimentary xenoliths, and metasedimentary rocks in the contact zone did have coherent paleodirections. In subsequent sampling xenoliths were preferentially selected. Magnetization directions observed at four sites were scattered, and demagnetization yielded no components that were consistent from specimen to specimen. Acceptable data was obtained at 13 of 17 sites sampled.

Two examples are described. At site 1 there is, after removal of a small soft magnetization, only a single, high unblocking temperature magnetization (Figure 2). The direction of these have been obtained by cleaning at a single field (in the range 20-40 mT) or temperature usually 500°C. Such simple signatures, however, are not common. A more typical result is that from site 10 at which unblocking temperatures are distributed throughout

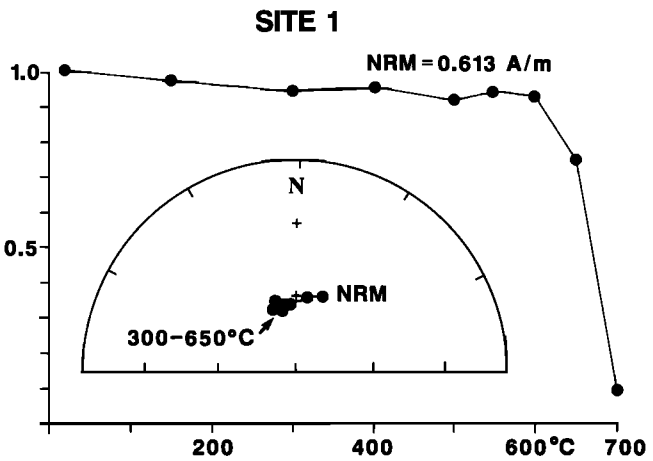


Fig. 2. Thermal demagnetization of single-component magnetization in metasedimentary xenolith, site 1 of Skelly Creek batholith. Intensity and direction changes are shown relative to present horizontal.

the range from room temperatures to 700°C, and at high temperatures, the directions, although scattered, become reversed in polarity (Figure 3). In the orthogonal diagram, the decay line is somewhat irregular and bypasses the origin. The principal magnetization is inclined downward and to the north and is essentially parallel to the single-

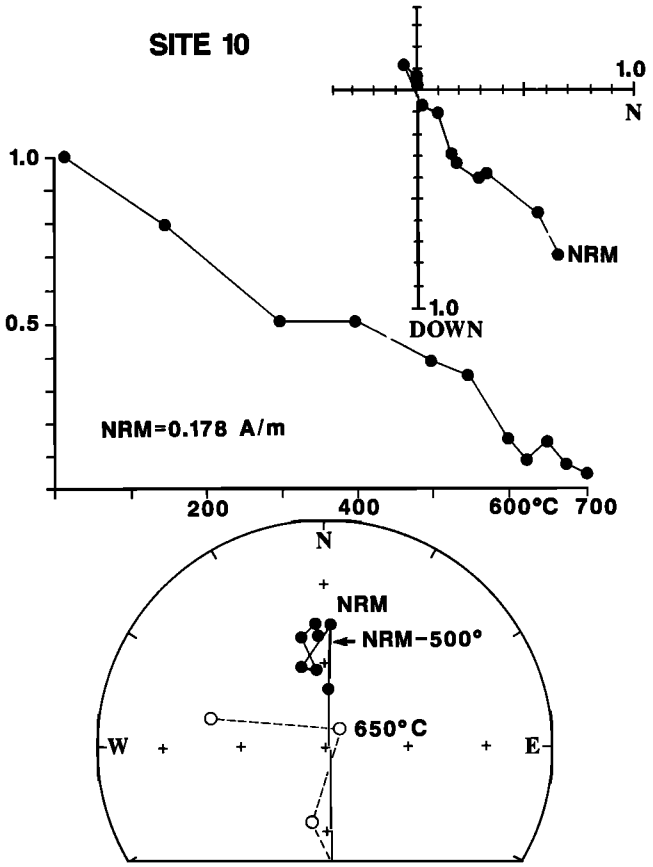


Fig. 3. Thermal demagnetization of two-component magnetization in granodiorite at site 10 of Skelly Creek batholith. Intensity, direction changes, and orthogonal plot in the N/S vertical plane are shown. Solid (open) symbols on the stereogram denote positive (negative) inclinations.

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

component magnetization at site 1. Most specimens have a similar complex character, and their principal magnetizations have been obtained by line fitting in the ranges of alternating fields and temperature listed in Table 1, using the procedures of *Zijderveld* [1967] and *Kirschvink* [1980].

The small high-temperature magnetizations are not well-defined. They often occur at sites which showed visible reddening along fine fractures, and we ascribe them to deposition of hematite during an interval of fracturing and fluid flow perhaps related in time to the massive extensional event observed immediately to the west [*Parrish et al.*, 1988]. The fact that the polarity of this "hematite" magnetization is commonly reversed in polarity indicates that the reddening is unlikely to have been caused by recent weathering.

Specimen directions are plotted in Figure 4, and mean site directions are listed in Table 1. Within-site precisions are variable, but the grouping of site means is good (k = 53), and approximates to that expected from mid-Cretaceous secular variation at these latitudes [*Globerman and Irving*, 1988]; it seems probable that the paleosecular variation has been averaged out. The paleopole is at 78°N,144°W (dm,dp = 10,9).

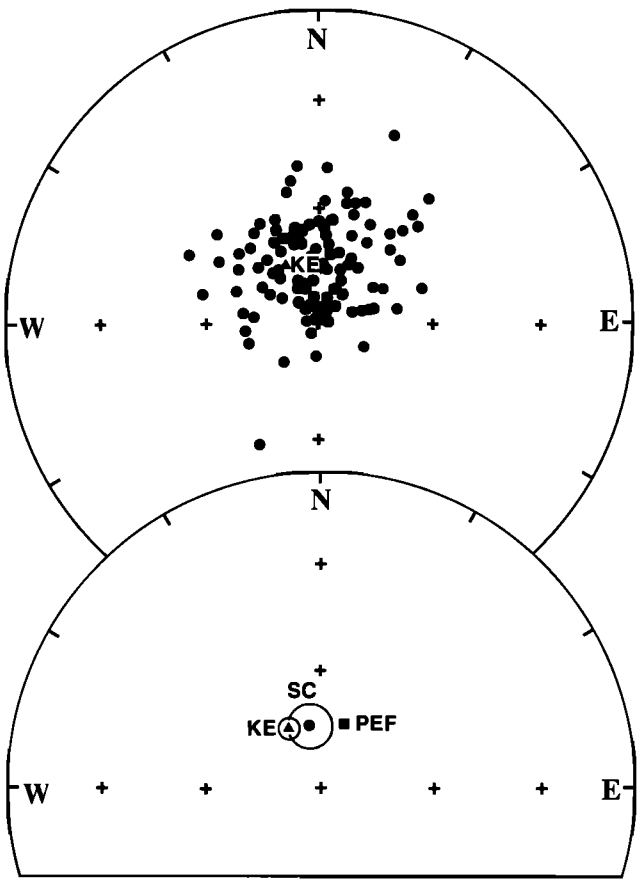


Fig. 4. Above, directions of principal magnetization in all specimens from the Skelly Creek batholith. Below, overall mean (SC) is compared with that expected (KE) for mid-Cretaceous time (Table 1). PEF is the present geomagnetic field direction. Standard error circles (P = 0.63) are shown. Directions all with respect to present horizontal.

The overall mean direction is not significantly different from that expected from the craton (Table 1 and Figure 4). The tilt required to bring the observed and expected directions together is 6° in a direction 277°E of north with an error of 6° (P = 0.05). This is consistent with the tilt determined from bathozonal data. Both bathozonal and paleomagnetic data indicate that tilting was small or negligible and that during the mid-Cretaceous the Purcell Anticlinorium was approximately in its present position relative to cratonic North America.

POSTEMPLACEMENT TILT OF SUMMIT STOCK

The Summit stock is a small, postkinematic pluton emplaced into west facing strata of the Late Proterozoic Quartzite Range Formation in the southern Kootenay Arc. Typically, it is leucocratic, medium- to coarse-grained, equigranular muscovite-bearing biotite monzogranite. Near the western contact it is a medium-grained granodiorite, has no primary muscovite, and contains abundant mafic xenoliths. Concordant biotite and muscovite dates of 102 Ma indicate a mid-Cretaceous age [*Archibald et al.*, 1983].

The stock is located approximately halfway between the faulted boundary which separates miogeoclinal strata and Jurassic volcanic rocks of the Rossland Group to the west and the Purcell Trench (Kootenay Lake) to the east. The Purcell Trench is the locus of a narrow, amphibolite facies, regional metamorphic culmination with a north-south orientation that transects the Kootenay Arc. West of the Purcell Trench, the metamorphic grade decreases, and the Summit stock was emplaced into greenschist facies rock [*Glover*, 1978]. Although in, and north of, the Summit Creek area, regional metamorphism and penetrative deformation are known to be mid-Jurassic in age [*Archibald et al.*, 1983], U-Pb dating of synkinematic plutons in the Corn Creek area (immediately east of the Summit Creek area) reveals that deformation and regional metamorphism were mid-Cretaceous in age (94 Ma [*Archibald et al.*, 1984]).

In the more pelitic members of the Lower Purcell Supergroup in the Corn Creek area, kyanite and sillimanite coexist [*Archibald et al.*, 1984]. Recently, G. Leclair (personal communication, 1988) found the assemblage, biotite-garnet-kyanite-sillimanite-staurolite, a few kilometers west of the Purcell Trench. These assemblages indicate pressure conditions at the time of metamorphism of 4-5.5 kbar (the bathozone 4 to 5 transition; see Figure 6 of *Archibald et al.* [1983]). In contrast, in the aureole of the Summit stock itself, the assemblage, quartz-muscovite-sillimanite-andalusite-K-feldspar is developed in younger strata, indicating emplacement at a lesser depth corresponding to the bathozone 1 to 2 transition (about 2.2 kbar [*Glover*, 1978]). Evidently, the Summit Creek and Corn Creek areas were at different crustal levels in mid-Cretaceous time. About 12 km of differential uplift and erosion in post-mid-Cretaceous time is implied.

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Results by Sites

Site	C(s)	M	D, I deg	k	α95 deg	N, E minutes arc
			Skelly Creek Batholith (49.4°N,116.6°W)			
1	9(17)	28.08	010,80	85	4	19.8,29.8
2	5 (8)	1.13	336,63	38	9	19.8,32.0
3	8(15)	0.26	018,80	19	9	24.6,28.1
4	6 (7)	0.28	304,78	11	19	20.7,30.8
5	6 (7)	0.13	347,66	28	12	21.2,28.3
10	8(14)	0.20	029,63	47	6	22.4,37.9
11	4 (5)	0.11	033,79	64	10	22.4,37.5
12	3 (6)	0.23	029,70	21	15	19.8,42.4
21	5(10)	0.31	319,77	39	8	19.6,32.6
22	3 (4)	0.10	309,69	293	5	19.7,32.6
23	3 (4)	0.25	356,71	248	12	19.8,29.9
24	3 (4)	0.08	333,71	62	9	23.0,37.2
29	6 (6)	0.11	330,64	21	13	23.2,40.5
Mean 13 sites			349,74	53	6	
			Summit Stock (49.1°N,117.0°W)			
51	8(14)	0.08	55,70	126	4	3.8, 3.7
52	6(12)	0.08	42,74	65	5	3.9, 4.0
53	3 (7)	0.20	39,75	161	5	3.9, 2.3
54	3 (5)	0.11	37,75	204	5	3.9, 2.3
55	3 (6)	0.15	62,70	121	6	3.8, 2.3
61	5(10)	0.47	46,63	34	8	3.8, 3.6
62	5 (9)	0.41	38,74	58	7	3.7, 3.5
63	4 (7)	0.22	53,74	46	9	3.8, 2.4
64	7(12)	0.68	54,74	46	9	5.0, 1.2
Mean (SSU) 9			48,72	290	3	
Mean (SSC) 9			318,74	290	3	
Expected -			332,74	-	3	

The site numbers are field accession numbers and have no geological significance. *C(s)* number of oriented cores and specimens cut from them, *M* mean intensity of magnetization (A/m), *D,I* declination and inclination of site-mean directions relative to present horizontal, *k* precision, α95 error ($P = 0.05$). N,E are the longitude (add 116°W for Skelly Creek batholith, 117°W for Summit stock) and latitude (add 49°N) in minutes. Rock types and treatment for Skelly Creek batholith are as follows: site 1, sedimentary xenoliths in granodiorite near contact, 40 mT, 500°C; 2, mafic xenoliths and granodiorite, 20 mT, 500°C; 3, mafic xenoliths, 5-20 mT, 150°-600°C; 4, mafic xenoliths, 10-25 mT, 150°-600°C; 5, mafic xenoliths, 10-20mT, 150°-600°C; 10, mafic xenoliths and granodiorite, 10-80 mT, 300°-600°C; 11, mafic xenoliths, 10-80 mT, 600°C; 12, mafic xenoliths, 10-60 mT, 150-600°C; 21, mafic xenoliths, 10-40 mT, 150°-500°C; 22, mafic xenoliths, 60 mT, 550°C; 23, metasediments near contact, 50 mT, 600°C; 24, mafic xenoliths, 30 mT, 500°C; 29, mafic xenoliths and granodiorite, 10-40 mT. The paleopole from the Skelly Creek batholith is at 78°N,144°W (dm=10, dp=9°). Rock types and treatments used for Summit stock are as follows: site 51, mafic xenolith and lamprophyre dike, 20 mT, 500°C; 52, mafic xenoliths, 20 mT, 500°C; 53,54,54, mafic xenoliths, 60 mT, 500°C; 61,62 massive granodiorite, line fits 10-100 mT; 63, mafic xenoliths, 30 mT; and site 64, massive diorite, 30 mT. Mean (SSU) is the average of nine sites in situ. Mean (SSC) is SSU corrected for tilt of 24° to the west as determined from bathozonal data. The Summit stock paleopole calculated from SSC is 64°N,166°W (dm=6°, dp=5°). The expected direction is calculated from 71°N,164°W, the mean cratonic reference paleopole for the interval 136, 100 Ma [*Globerman and Irving*, 1988].

There is no geological evidence to suggest anything but a smooth transition from high-pressure conditions in the east to low-pressure conditions in the west. The separation of the occurrences of the bathozone 1 to 2 and bathozone 4 to 5 transitions is about 26 km. Thus, using the pressure estimates mentioned above, an error of 0.5 kbar in the bathozone 1 to 2 transition, an error of 1.0 kbar in the bathozone 4 to 5 transition, and a conversion factor of 3.7 km/kbar, the calculated tilt of the mid-Cretaceous paleohorizontal across this region is 24°±10° down to the west.

PALEOMAGNETISM OF THE SUMMIT STOCK

Cores were collected from nine sites distributed along highway 3 about 43 km west of Creston, British Columbia (Figure 1). The granodiorite itself was sampled, but most samples were from mafic xenoliths which, as in the Skelly Creek batholith, were found to be more stable magnetically. A lamprophyre dike was also sampled.

The magnetizations were single component, with high coercivities (commonly 30% of the remanence remained after demagnetization in 100 mT), and

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

high unblocking temperatures (500-650°C). Directions remained little changed during demagnetization. Orthogonal plots show linear decay to the origin, and directions are unchanged up to 600°C (Figure 5). Most specimens (mafic xenoliths, massive diorite, lamprophyre) behaved in this fashion, and they were "cleaned" at a single temperature or demagnetizing field. The granodiorites of sites 61 and 62 gave decay lines that converged on the origin but were less well defined. Their magnetizations were obtained by least squares fit of decay lines obtained in the demagnetization intervals given in Table 1.

Individual specimen directions are very well grouped (Figure 6). The demagnetization conditions, site mean directions, and other details are given in Table 1. The overall mean direction (SSU) relative to present-day horizontal is (*D,I*) 48°,72°, k = 290, α_{95} = 3.0°, N (sites) = 9, and the paleopole situated at 61°N,62°W, dm,dp = 5°,5°.

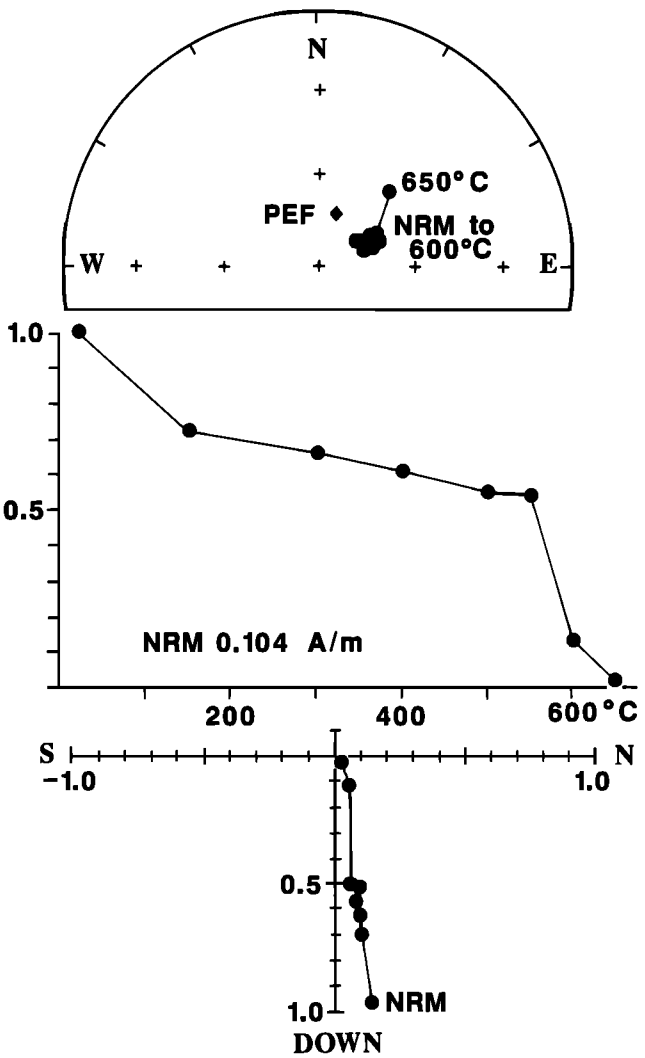


Fig. 5. Thermal demagnetization of diorite xenolith from site 52 of Summit stock. Top, direction changes relate to present horizontal. Middle, intensity changes. Bottom, normalized orthogonal plot (N-S vertical plane).

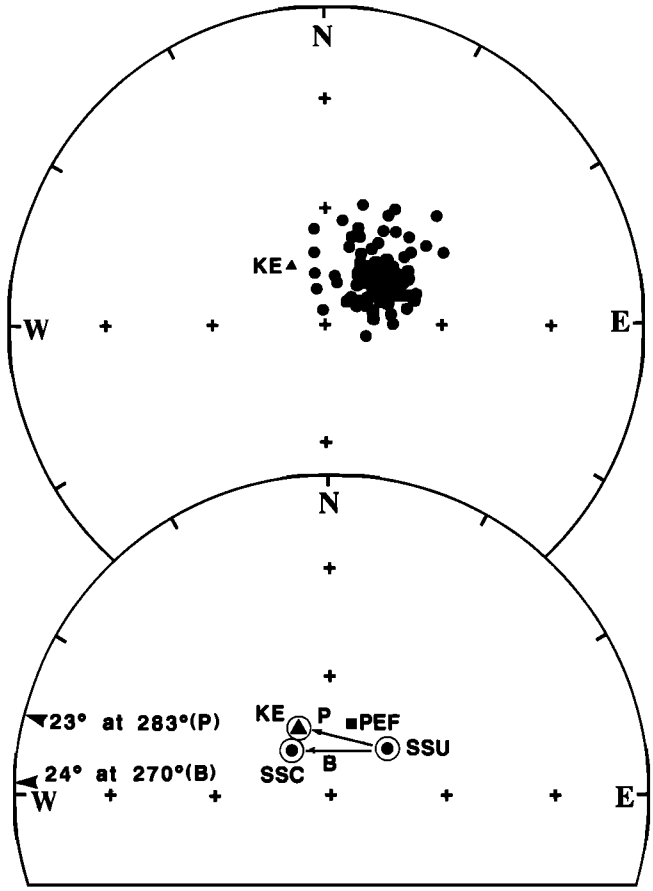


Fig. 6. Top, Summit stock specimen directions relative to present horizontal. Bottom, mean direction compared with (KE) the expected Cretaceous paleofield (Table 1). SSU mean direction relative to present horizontal, SSC corrected for tilt determined from bathozones (Table 1). The arrows labelled B and P denote tilts estimated from bathozonal and paleomagnetic data respectively. PEF is the present geomagnetic field direction. Standard error circles (P = 0.63) are given.

The direction relative to paleohorizontal, using a tilt of 24° to the west as estimated from bathozonal data, is 318°,74°, and the paleopole is at 64°N,166°W.

Other than that it must be post-mid-Cretaceous, data from the Summit stock itself provide no constraints on the age of the westerly tilt proposed herein. Recently, however, westerly tilts of 28° and 38°, respectively, have been observed from the Eocene Ymir stock (52 Ma [*Archibald et al.*, 1983]) 25 km to the northwest and from Eocene core-complex rocks of Lower Arrow Lake 50 km to the west northwest [*Bardoux and Irving*, 1989], tilts which arise presumably from Eocene extension. Latest Cretaceous and Paleocene Rb-Sr muscovite dates and Paleocene to Eocene K/Ar mica dates suggest that the rocks immediately west of the Purcell Trench remained hot long after mid-Cretaceous time [*Archibald et al.*, 1984]. Thus it is probable that the tilt of the Summit stock also resulted from Eocene extension occurring at the same time as uplift, erosion, and cooling of high-grade metamorphic rocks exposed along the Purcell Trench.

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

After correction for tilt, the paleodirection is in excellent agreement with that from cratonic North America (Table 1 and Figure 6). Conversely, the tilt required to bring the mean Summit stock paleodirection into agreement with that expected from the craton is 23° at 283°E of north, which is in good accord with the bathozonal estimate.

It could be argued that this agreement between observed and expected values is coincidental, the cooling time represented by the sampling sites being too short to average out the paleosecular variation. It is likely that the precision of site mean directions (k = 290) is too high to represent fully the paleosecular variation. Alternatively, because cooling is likely to have been slow, each site may represent an appreciable time average of the paleofield. This uncertainty renders the discussion of the Summit stock data less than fully satisfactory, but we argue that for a statistically inadequate average to have produced the remarkable agreement of Figure 6 makes appeal to coincidences beyond what may be considered reasonable. For example, the observed and expected inclinations agree to the nearest degree. Hence we conclude that results from both the Summit stock and the Skelly Creek batholith indicate that this part of the eastern Omineca Belt lay close to the margin of ancestral North America in the mid-Cretaceous, very much in the position it occupies today, a conclusion that is consistent with geological interpretations. Evidently, aberrant magnetization in mid-Cretaceous rocks of the Cordillera does not extend as far east as the eastern part of the Omineca Belt.

RELEVANCE TO TERRANE TECTONICS

Rocks of the miogeocline are thrust to the west over the Jurassic volcanics of the Rossland Group which are part of eastern Quesnellia. Quesnellia is a terrane which, to the west, occupies the Intermontane Belt and to the east the western part of the Omineca Belt. The compressional event which brought Quesnellia and the eastern part of the Omineca Belt together and which is evidenced by this westerly directed thrusting, is probably mid-Jurassic in age. It certainly occurred before the mid-Cretaceous, because associated thrust faults are "plugged" by mid-Cretaceous plutons [*Archibald et al.*, 1984]. If these geological inferences are correct, then the paleomagnetic data from the Skelly Creek batholith and Summit stock indicate that the eastern portion of Quesnellia also was close to its present position relative to the North American craton during mid-Cretaceous time.

Acknowledgments. Thanks are due to Judith Baker, who made the magnetic measurements; to Jane Wynne for help with computations; to M.E. Beck, Jr., C.R. Yorath, J.W. Hillhouse, and an anonymous reviewer for their very helpful comments. Fieldwork, metamorphic studies, and geochronology, were supported by an EMR research agreement with D.A. Archibald and E. Farrar and an NSERC grant to E. Farrar. Contribution no. 16689 Geological Survey of Canada; LITHOPROBE contribution no. 102.

REFERENCES

Archibald, D.A., J.K. Glover, R.A. Price, E. Farrar, and D.M. Carmichael, Geochronology and tectonic implications of magmatism and metamorphism, southern Kootenay Arc and neighbouring regions, southeastern British Columbia, part I, Jurassic to mid-Cretaceous, *Can. J. Earth Sci., 20,* 891-1913, 1983.

Archibald, D.A., T.E. Krogh, R.L. Armstrong, and E. Farrar, Geochronology and tectonic implications of magmatism and metamorphism, southern Kootenay Arc and neighbouring regions, southeastern British Columbia, part II, mid-Cretaceous to Eocene, *Can. J. Earth Sci., 21,* 567-583, 1984.

Armstrong, R.L., J.W.H. Monger, and E. Irving, Age of magnetization of the Axelgold gabbro, north-central British Columbia, *Can. J. Earth Sci., 22,* 1217-1222, 1985.

Bardoux, M. and E. Irving, Paleomagnetism of Eocene rocks of the Kelowna and Castlegar areas, British Columbia: Studies in determining paleohorizontal, *Can. J. Earth Sci., 26,* 829-844, 1989.

Beck, M.E., Jr., and L. Noson, Anomalous paleolatitude in Cretaceous granitic rocks, *Nature, 235,* 11-13, 1972.

Beck, M.E., Jr., R.F. Burmester, and R. Schoonover, Paleomagnetism and tectonics of the Cretaceous Mt. Stuart batholith of Washington: Translation or tilt? *Earth Planet. Sci. Lett., 56,* 336-342, 1981.

Carmichael, D.M., Metamorphic bathozones and bathograds: A measure of the depth of post-metamorphic uplift and erosion on the regional scale, *Am. J. Sci., 278,* 769-797, 1978.

Gabrielse, H., Major dextral transcurrent displacements along the northern Rocky Mountain Trench and related lineaments in north-central British Columbia, *Geol. Soc. Am. Bull., 96,* 1-14, 1985.

Globerman, B.R., and E. Irving, Mid-Cretaceous paleomagnetic reference field for North America: Restudy of 100 Ma intrusive rocks from Arkansas and overprints in older rocks, *J. Geophys. Res., 93,* 11,721-11,734, 1988.

Glover, J.K., Geology of Summit Creek Map area, southern Kootenay Arc, British Columbia, Ph.D. thesis, 144 pp., Queen's Univ. Kingston, Ont. 1978.

Hoy, T., and P. van der Heyden, Geochronological studies in the East Kootenay region, *Geological Field-Work 1988,* British Columbia Dept. Energy, Mines and Petrol. Res., vol. 1, pp. 25-36, 1988.

Irving E., and N.W.D. Massey, Paleomagnetism of ocean-layers 2 and 3, evidence from the Metchosin Complex, Vancouver Island, *Phys. Earth Planet. Inter.,* in press, 1989.

Irving, E., G.J. Woodsworth, P.J. Wynne, and A. Morrison., Paleomagnetic evidence for displacement from the south of the Coast Plutonic Complex, British Columbia, *Can. J. Earth Sci., 22,* 1217-1222, 1985.

Kirschvinck, J.L., The least-square line and plane and the analysis of paleomagnetic data, *Geophys. J. R. Astron. Soc., 62,* 699-718, 1980.

Marquis, G., and B.R. Globerman, Northward motion of the Whitehorse Trough: paleomagnetic evidence from the Upper Cretaceous Carmacks Group, *Can. J. Earth Sci., 25,* 2005-2016, 1988.

Monger, J.K.H., and E. Irving, Northward displacement of north-central British Columbia, *Nature, 288,* 289-294, 1980.

Parrish, R., Carr, S.D. and Parkinson, D.L., Eocene extensional tectonics and geochronology of the southern Omineca Belt, British Columbia and Washington, *Tectonics, 7,* 181-212, 1988.

Price, R.A., The Cordilleran foreland thrust and fold belt in the southern Canadian Rocky Mountains, Thrust and Nappe Tectonics, edited by N.J. Price and K. MacClay, *Geol. Soc. London Spec. Pub. 9,* 427-448, 1981.

Price, R.A., and D.M. Carmichael, Geometric test for Late Cretaceous-Paleogene intracontinental transform faulting in the Canadian Cordillera, *Geology, 14,* 468-471, 1986.

Ressor, J.E., Dewar Creek map-area with special emphasis on the White Creek batholith, B.C., *Geol, Surv. Can. Mem., 292,* 1-78, 1958.

Symons, D.T.A., Geotectonics of Cretaceous and Eocene plutons in British Columbia, a paleomagnetic fold test, *Can. J. Earth Sci., 14,* 1246-1262, 1977.

Zidjerveld, J.D.A., A.C. demagnetization of rocks: Analysis of results, in *Methods in Palaeomagnetism,* edited by D.W. Collison and S.K. Runcorn, pp. 254-286, Elsevier, New York, 1967.

D.A. Archibald, Geological Sciences, Queen's University, Kingston, Ontario, Canada K7L 3N6.

E. Irving, Pacific Geoscience Centre, Geological Survey of Canada, Sidney, British Columbia, Canada V8L 4B2.

(Received June 13, 1989;
revised September 13, 1989;
accepted September 20, 1989.)

21562202b, 1990, B4, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JB095iB04p04579 by Curtin University Library, Wiley Online Library on [05/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License